

Almond Yield as a Function

Ixel Medrano

Sofia Sarak

ASSIGNMENT 3 Almond Yields

Lobell et al., 2006 used data to build models of tree crop yield for California that captured how climate variation (place and time) might influence yield

- Yield anomaly
- Climate variables

Assumptions...as you work think about what these are...

Assignment

For this assignment you will work in *pairs*

Your goal is to implement a simple model of almond yield anomaly response to climate

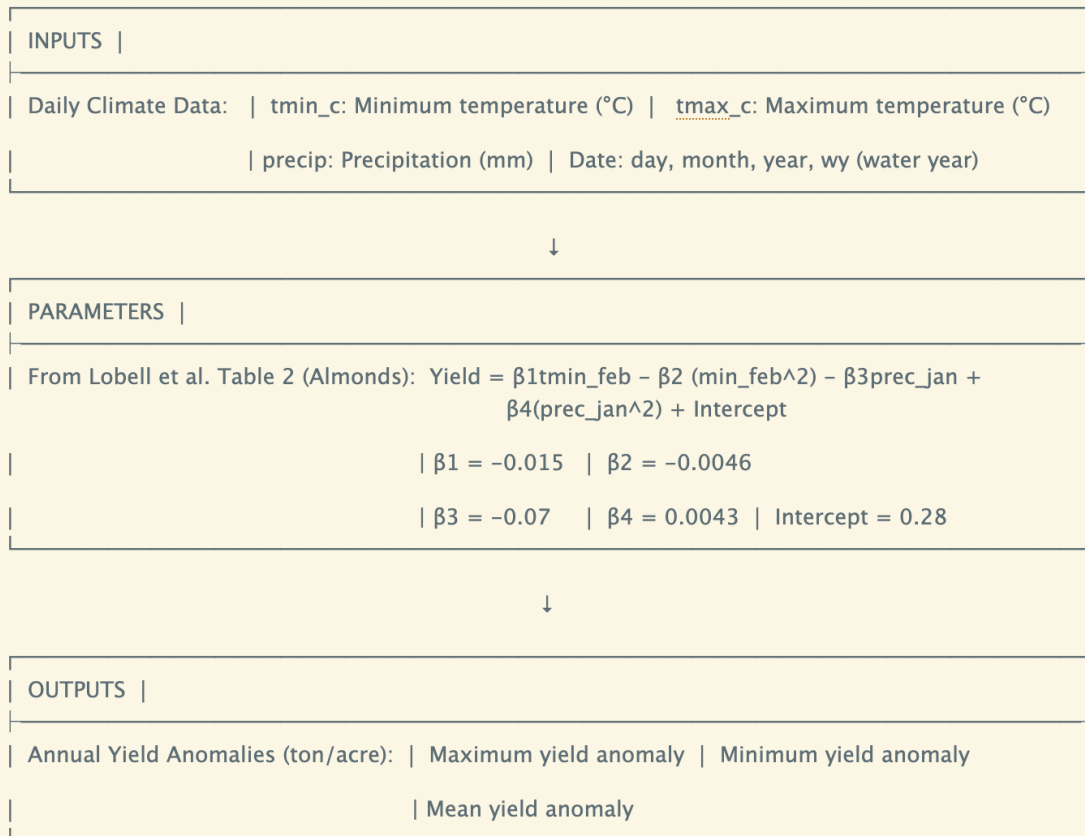
- Inputs: daily times series of minimum, maximum daily temperatures and precipitation
- Outputs: maximum, minimum and mean yield anomaly for a input time series

The Lobell et al. 2006 paper will be the source for the transfer function that you will use in your model; specifically look at the equations in table 2.

What you will do - Model Set Up

1. Draw diagram to represent your model - how it will translate inputs to outputs, with parameters that shape the relationship between inputs and outputs - on your diagram list what your inputs, parameters and outputs are with units

Conceptual Model Diagram



2. Implement your diagram as an R function

```
# Load necessary libraries
library(tidyverse)
library(here)
library(janitor)

# Read in data
```

```
climate <- read_table(here("week 3/clin.txt")) %>%
  clean_names()
```

Here are some ideas to think though. The climate data is going to need to be processed (e.g. note that the model uses climate data from a particular month - you have ALL of the climate as daily data). Here are two possible model outlines to follow

```
Almond_model <- function(clim_var1, clim_var2, parameters){.....}
```

```
Almond_model <- function(clim, parameters){.....}
```

The first example is where the climate variables are separately input into the function - climate data is processed beforehand as part of model set up

The second is where a climate data frame is the input to the function and you extract the useful data from it as part of the model.

```
# Initialize function
alm_yield <- function(climate_data){

  # Find mean minimum temperature in February of each year
  climate_month <- climate_data %>%
    filter(month == 2) %>%
    group_by(year) %>%
    summarize(min_temp_feb = mean(tmin_c))

  # Find total precipitation in January of each year
  precip_month <- climate_data %>%
    filter(month == 1) %>%
    group_by(year) %>%
    summarize(precip_sum_jan = sum(precip))

  # Extract necessary variables from respective dfs
  tmin_feb <- climate_month$min_temp_feb
  prec_jan <- precip_month$precip_sum_jan

  # Apply formula to variables
  yield <- (-0.015 * tmin_feb) - (0.0046 * (tmin_feb)^2) - (0.07 * prec_jan) + (0.0043 * (pr

  # Print resulting yield
  print(yield)
}
```

You can pick which option you prefer (or try both) If you want a coding challenge, you could create a model that works for any crop

Run the model

3. Run your model for the **clim.txt** data that is posted on Canvas,

clim.txt has the following columns

these 4 columns tell you when climate observations were made

- *day*
- *month*
- *year*
- *wy* (water year)

the next 3 columns are the climate observations

- maximum (*tmax_c*) daily temperature
- minimum (*tmin_c*) daily temperature
- precipitation (*precip*) on that day in mm

```
# Running function on climate data
alm_yield(climate_data = climate)
```

```
[1] -0.3552237  9.2906757  68.9130633  15.4280698  20.2083803
[6]  2.4820009 1919.9811511   3.5818399 329.6938750  27.8636956
[11] -0.1436364   9.5999883 159.5119587   0.2450914 -0.2585997
[16] -0.2367722  656.3724121  18.6324135  20.2007396 576.2821943
[21]  0.7367438 153.7655092
```

Check your work

I will tell you that

- the maximum almond yield anomaly should be approximately 1920 ton/acre
- the lowest almond yield anomaly should be approximately -0.027 ton/acre
- the mean almond yield anomaly should be approximately 182 ton/acre

```
[1] "Min almond yield: -0.36"
```

```
[1] "Max almond yield: 1919.98"
```

```
[1] "Mean almond yield: 181.45"
```